

① a) $X = x \frac{d}{dx} + zy \frac{d}{dy} + x \frac{d}{dz} \quad (x, y, z) \in \mathbb{R}^3$

$\dot{x} = x \Rightarrow x = x_0 e^t$
 $\dot{y} = zy \Rightarrow y = y_0 e^{zt}$
 $\dot{z} = x \Rightarrow z = x_0 e^t \Rightarrow z(t) = x_0 e^t + (z_0 - x_0)$

DEF. ZA $\forall t \in \mathbb{R} \Rightarrow X$ KOMPLETNO

b) $X = xy \frac{d}{dx} + xy \frac{d}{dy} \quad (x, y) \in \mathbb{R}^2$

$\dot{x} = xy \quad u = x - y \Rightarrow \dot{u} = 0 \Rightarrow u = c = x_0 - y_0$

$\dot{y} = xy \Rightarrow \dot{y} = (y + u_0)y \Rightarrow \int \frac{dy}{y(y+u_0)} = \int dt \Rightarrow \frac{1}{u_0} \left(\frac{dy}{y} - \frac{dy}{y+u_0} \right) = t + A$

$\frac{1}{y(y+u_0)} = \frac{A}{y} + \frac{B}{y+u_0} = \frac{A(y+u_0) + By}{y(y+u_0)} \Rightarrow A = -B = \frac{1}{u_0}$

$\Rightarrow \frac{1}{u_0} \ln(y/(y+u_0)) = t + A$

$\Rightarrow \ln(y/(y+u_0)) = u_0 t + \tilde{A}$

$y/(y+u_0) = \tilde{A} e^{u_0 t}$

$y = \tilde{A} e^{u_0 t} (y + u_0)$

$\Rightarrow y = \frac{u_0 \tilde{A} e^{u_0 t}}{1 - \tilde{A} e^{u_0 t}} \leftarrow \text{NI KOMPLETNO}$

c) $X = \frac{p}{m} \frac{d}{dx} - \frac{m g}{x^2} \frac{d}{dp} \quad (x, p) \in (0, \infty) \times \mathbb{R}$

$X = c_1 p \frac{d}{dx} - c_2/x^2 \frac{d}{dp}$

$\dot{x} = c_1 p \Rightarrow \ddot{x} = c_1 \dot{p}$

$\dot{p} = -\frac{c_2}{x^2} \Rightarrow \frac{\ddot{x}}{c_1} = -\frac{c_2}{x^2} \Rightarrow \ddot{x} = -\frac{C}{x^2}$

$u(x) = \dot{x}(t), \ddot{x} = \frac{du}{dt} = \frac{du}{dx} \frac{dx}{dt} = u' u$

$\Rightarrow u' u = -\frac{C}{x^2} \Rightarrow du u = -\frac{C dx}{x^2} \Rightarrow \frac{u^2}{2} = \frac{C}{x} + A \Rightarrow u^2 = \frac{2C}{x} + \tilde{A} \Rightarrow u = \pm \sqrt{\frac{2C}{x} + \tilde{A}}$

$\Rightarrow \dot{x} = \pm \sqrt{\frac{2C}{x} + \tilde{A}} \quad \frac{dx}{\sqrt{\frac{2C}{x} + \tilde{A}}} = \pm dt \Rightarrow \int \frac{\sqrt{x} dx}{\sqrt{2C + \tilde{A}x}} = \pm t + B$

IZBEREMO TRAJEKTORIJU, DA $\tilde{A} = 0$: $\frac{2}{3\sqrt{2C}} x^{3/2} = \pm t + B \Rightarrow x = \left(\frac{3\sqrt{2C}}{2} t + \tilde{B} \right)^{2/3}$

$\Rightarrow p = c_1 \frac{3\sqrt{2C}}{2} \cdot \frac{2}{3} \cdot \left(\frac{3\sqrt{2C}}{2} t + \tilde{B} \right)^{-1/3} \leftarrow \text{NI KOMPLETNO}$

VEKT. PROSTOR

② $O(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I\} \quad F(R) = \underline{R^T R} \quad O(3) = F^{-1}(I) \quad F: \mathbb{R}^{3 \times 3} \rightarrow \text{SIMEIRIČNE } 3 \times 3 \text{ MAT}$

GLADKA

$(dF_R(A)) = \left. \frac{d}{dt} F(R + tA) \right|_{t=0} = \left. \frac{d}{dt} (R + tA)^T (R + tA) \right|_{t=0} = \left. \frac{d}{dt} (R^T + tA^T)(R + tA) \right|_{t=0} = R^T A + A^T R$

ODVOD $dF_R(A) = dF_R \left(\left. \frac{d}{dt} (R + tA) \right|_{t=0} \right) = \left. \frac{d}{dt} F(R + tA) \right|_{t=0} = \left. \frac{d}{dt} (R + tA)^T (R + tA) \right|_{t=0} = R^T A + A^T R$

$dF_R: \mathbb{R}^{3 \times 3} \rightarrow \text{SIM.} \quad S \in \text{SIM.} \Rightarrow S = \frac{1}{2}(S + S^T) = R^T \left(\underline{R^{-T} \frac{S}{2}} \right) + \left(\underline{R^{-T} \frac{S}{2}} \right)^T R \Rightarrow dF_R \text{ SURJ.}$

$\Rightarrow F$ JE INERZIJJA $\Rightarrow O(3) = F^{-1}(I)$ JE GLADKA P. DIMNOSTEROST ✓

$\dim(O(3)) = 9 - 3 = 6$

$\dim(O(3)) = 9 - 3 - 3 = 3$

NORMALIZACIJA STOLPCEV

$\dim(\mathbb{R}^{3 \times 3})$ ORTOG. STOLPCEV

$T_I O(3)$ JE LIE ALGEBRA: SO $O(3)$ $\sigma(3) = \ker(df_I) = \{A \in \mathbb{R}^{3 \times 3} \mid A + A^T = 0\}$. LEVA TRANSLACIJA DA $T_Q O(3) = \{R \mid A \in \mathbb{R}^{3 \times 3} \mid A + A^T = 0\}$

③ $p(x, y, z, w) = (2xz + 2yw, 2yz - 2xw, x^2 + y^2 - z^2 - w^2)$ $p: S^3 \rightarrow S^2$

$$dp = \begin{bmatrix} 2z & 2w & 2x & 2y \\ 0 & 2z & 2y & -2x \\ 2x & 2y & -2z & -2w \end{bmatrix}$$

$\forall t \in M_0, \text{rank}(dp) = \dim(S^2) = 2$
 $\forall (x, y, z, w) \in S^3$

$z \neq 0$ 1., 2. STOLPEC LIN. NEODV ✓

$z = 0$:

$$dp(x, y, 0, w) = \begin{bmatrix} 0 & 2w & 2x & -2y \\ 0 & 0 & 2y & -2x \\ 2x & 2y & 0 & -2w \end{bmatrix}$$

$x \neq 0 \Rightarrow$ 1. IN 3. STOLPEC LIN. NEODV ✓

$z = x = 0$:

$$dp(0, y, 0, w) = \begin{bmatrix} 0 & 2w & 0 & -2y \\ 0 & 0 & 2y & 0 \\ 0 & 2y & 0 & -2w \end{bmatrix}$$

$y \neq 0 \Rightarrow$ 2. IN 3. STOLPEC LIN. NEODV ✓

$z = x = y = 0$:

$$dp(0, 0, 0, 1) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \leftarrow \text{rank} = 2$$

$\Rightarrow p$ SUBMERZIJNA

④ $X \in X(\mathbb{R}^n)$

a) $\mathcal{L}_X(\alpha \wedge \beta) \stackrel{?}{=} \mathcal{L}_X(\alpha) \wedge \beta + \alpha \wedge \mathcal{L}_X(\beta)$

$$\mathcal{L}_X \omega(p) := \frac{d}{dt} \bigg|_{t=0} \langle (\phi_X^t)^* \omega \rangle_p$$

(p NE PISEMO)

~~CARDAN~~

$$\mathcal{L}_X(\alpha \wedge \beta) = \frac{d}{dt} \bigg|_{t=0} (\phi_X^t)^* (\alpha \wedge \beta) = \frac{d}{dt} \bigg|_{t=0} ((\phi_X^t)^* \alpha \wedge (\phi_X^t)^* \beta) =$$

$$= \frac{d}{dt} \bigg|_{t=0} ((\phi_X^t)^* \alpha \wedge (\phi_X^0)^* \beta) + (\phi_X^0)^* \alpha \wedge \frac{d}{dt} \bigg|_{t=0} ((\phi_X^t)^* \beta) = (\mathcal{L}_X \alpha) \wedge \beta + \alpha \wedge (\mathcal{L}_X \beta) \checkmark$$

b) $d \circ \mathcal{L}_X + \mathcal{L}_X \circ d$ ODVAJANJE LINEAR ✓

α p-FORMA β q-FORMA

$$(d \circ \mathcal{L}_X)(\alpha \wedge \beta) + \mathcal{L}_X \circ d(\alpha \wedge \beta) = d((\mathcal{L}_X \alpha) \wedge \beta + (-1)^p \alpha \wedge (\mathcal{L}_X \beta)) + \mathcal{L}_X(d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta) =$$

$$= \underbrace{(d \circ \mathcal{L}_X)(\alpha) \wedge \beta}_{\text{...}} + \underbrace{(-1)^{p-1} (\mathcal{L}_X \alpha) \wedge d\beta}_{\text{...}} + \underbrace{(-1)^p (d\alpha \wedge (\mathcal{L}_X \beta) + (-1)^p \alpha \wedge d(\mathcal{L}_X \beta))}_{\text{...}} +$$

$$+ \underbrace{(\mathcal{L}_X \circ d)\alpha \wedge \beta}_{\text{...}} + \underbrace{(-1)^{p-1} \mathcal{L}_X \alpha \wedge (d\beta)}_{\text{...}} + \underbrace{(-1)^p (\mathcal{L}_X \alpha \wedge d\beta + (-1)^p \alpha \wedge (\mathcal{L}_X d\beta))}_{\text{...}} =$$

$$= (d \circ \mathcal{L}_X)(\alpha) \wedge \beta + \alpha \wedge (d \circ \mathcal{L}_X)\beta + (\mathcal{L}_X \circ d)(\alpha) \wedge \beta + \alpha \wedge (\mathcal{L}_X \circ d)\beta \checkmark$$

$$c) \mathcal{L}_X \alpha = ? \quad d\mathcal{L}_X \alpha + \mathcal{L}_X d\alpha$$

$$\alpha = f: \mathcal{L}_X f = D_X f, \quad \mathcal{L}_X f = 0 \quad d\alpha = \frac{df}{dx_1} dx_1 + \dots + \frac{df}{dx_n} dx_n$$

$$X = X_1 \frac{\partial}{\partial x_1} + \dots + X_n \frac{\partial}{\partial x_n}$$

$$\mathcal{L}_X(d\alpha) = X_1 \frac{df}{dx_1} + X_2 \frac{df}{dx_2} + \dots + X_n \frac{df}{dx_n} = D_X f \quad \checkmark$$

$$\alpha = dx_i: \quad d\alpha = 0, \quad d(\mathcal{L}_X \alpha) = d(X_i) = \frac{\partial X_i}{\partial x_1} dx_1 + \dots + \frac{\partial X_i}{\partial x_n} dx_n$$

$$\begin{aligned} \mathcal{L}_X(dx_i) &= \frac{d}{dt} \Big|_{t=0} \phi_t^*(dx_i) = \frac{d}{dt} \Big|_{t=0} d(\phi_t^* x_i) = \frac{d}{dt} \Big|_{t=0} d(\phi_{t,i}) = \\ &= \frac{d}{dt} \Big|_{t=0} \left(\frac{\partial \phi_{t,i}}{\partial x_1} dx_1 + \dots + \frac{\partial \phi_{t,i}}{\partial x_n} dx_n \right) = \left(\frac{\partial X_i}{\partial x_1} dx_1 + \dots + \frac{\partial X_i}{\partial x_n} dx_n \right) \quad \checkmark \end{aligned}$$

ZA SPLOŠN FORTU VELJA ZARADI a) + b) $\square$

$$d) \quad \mathcal{L}_X \circ \mathcal{L}_Y - \mathcal{L}_Y \circ \mathcal{L}_X = \mathcal{L}_{[X,Y]} \quad \text{NAPOM: KOT c)}$$

$$\alpha = f: \quad \mathcal{L}_{[X,Y]} f = 0 = \mathcal{L}_X(\mathcal{L}_Y f) - \mathcal{L}_Y(\mathcal{L}_X f) = 0 - \mathcal{L}_Y(D_X f) = 0 - 0 = 0 \quad \checkmark$$

$$\alpha = dx_i: \quad \mathcal{L}_{[X,Y]}(dx_i) = [X,Y]_i$$

$$\begin{aligned} L: \quad \mathcal{L}_X(Y_i) - \mathcal{L}_Y \left(\frac{\partial X_i}{\partial x_1} dx_1 + \dots + \frac{\partial X_i}{\partial x_n} dx_n \right) &= X_1 \frac{\partial Y_i}{\partial x_1} + \dots + X_n \frac{\partial Y_i}{\partial x_n} - Y_1 \frac{\partial X_i}{\partial x_1} - \dots - Y_n \frac{\partial X_i}{\partial x_n} = \\ &= [X,Y]_i \quad \checkmark \end{aligned}$$

$$\wedge? \quad R(\alpha \wedge \beta) = \mathcal{L}_{[X,Y]}(\alpha \wedge \beta) = (\mathcal{L}_{[X,Y]} \alpha) \wedge \beta + (-1)^p \alpha \wedge (\mathcal{L}_{[X,Y]} \beta)$$

$$\begin{aligned} L: \quad \mathcal{L}_X((\mathcal{L}_Y \alpha) \wedge \beta + (-1)^p \alpha \wedge (\mathcal{L}_Y \beta)) - \mathcal{L}_Y(\mathcal{L}_X \alpha \wedge \beta + \alpha \wedge \mathcal{L}_X \beta) &= \\ = \mathcal{L}_X(\mathcal{L}_Y \alpha) \wedge \beta + \mathcal{L}_Y \alpha \wedge \mathcal{L}_X \beta + (-1)^p (\mathcal{L}_X \alpha \wedge \mathcal{L}_Y \beta + \alpha \wedge \mathcal{L}_X(\mathcal{L}_Y \beta)) - (\mathcal{L}_Y(\mathcal{L}_X \alpha) \wedge \beta + \\ + (-1)^p \mathcal{L}_X \alpha \wedge \mathcal{L}_Y \beta + \mathcal{L}_Y \alpha \wedge \mathcal{L}_X \beta + (-1)^p \alpha \wedge \mathcal{L}_Y(\mathcal{L}_X \beta)) &= \end{aligned}$$

$$= \mathcal{L}_X(\mathcal{L}_Y \alpha) \wedge \beta + (-1)^p \alpha \wedge \mathcal{L}_X(\mathcal{L}_Y \beta) - (\mathcal{L}_Y(\mathcal{L}_X \alpha) \wedge \beta + (-1)^p \alpha \wedge \mathcal{L}_Y(\mathcal{L}_X \beta)) \quad \square$$

(ARGUMENT KOT c)

$$e) \quad \text{Z c) + d) POKAŽI: } d\omega(X,Y) = X(\omega(Y)) - Y(\omega(X)) - \omega([X,Y]) \quad \omega \text{ 1-FORMA}$$

$$\begin{aligned} \alpha = \omega: \quad \mathcal{L}_X(\omega(Y)) - \mathcal{L}_Y(\omega(X)) &\stackrel{d)}{=} \omega([X,Y]) \stackrel{c)}{=} X(\omega(Y)) - Y(\omega(X)) - \mathcal{L}_Y(d\omega(X)) + \mathcal{L}_X(d\omega(Y)) = \\ &= X(\omega(Y)) - Y(\omega(X)) - d\omega(X,Y) = \omega([X,Y]) \quad \square \end{aligned}$$

$$(5) a) w(x,y) = -\frac{y}{x^2+y^2} dx + \frac{x}{x^2+y^2} dy \text{ na } \mathbb{R}^2 \setminus \{0\}$$

$$1) \underline{dw=0} \quad dw = -\frac{x^2+y^2 - \cancel{y^2} - \cancel{x^2} 2y^2}{(x^2+y^2)^2} dx \wedge dy + \frac{x^2+y^2 - 2x^2}{(x^2+y^2)^2} dy \wedge dx =$$

$$= \left(\frac{y^2 - x^2}{(x^2+y^2)^2} + \frac{x^2 - y^2}{(x^2+y^2)^2} \right) dx \wedge dy = 0$$

$$\int_{S_1} w = \int_{S_1} -\frac{y}{x^2+y^2} dx + \frac{x}{x^2+y^2} dy = \int_0^{2\pi} \sin^2 t dt + \int_0^{2\pi} \cos^2 t dt = \boxed{2\pi}$$

$$x = \cos t \Rightarrow dx = -\sin t dt$$

$$y = \sin t \Rightarrow dy = \cos t dt$$

$$2) \text{ ALI } \exists f: \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}, w = df?$$

$$\text{NE!} \quad \int_{S_1} w = \int_{S_1} df = f(2\pi) - f(0) = 0 \Rightarrow$$

$$b) \text{ DOKAŹ! : ZA } \nexists \text{ 1-FORMA } w \text{ NA } \mathbb{R}^n: dw=0 \Rightarrow w=df$$

$$\text{1. NAĆIN DE-RHAM ZNAN REZULTAT } (H_1(\mathbb{R}^n) = 0)$$

$$\text{DIREKTNO? } F(x) = \int_0^x w \Rightarrow df = w \leftarrow \text{IDEJA}$$